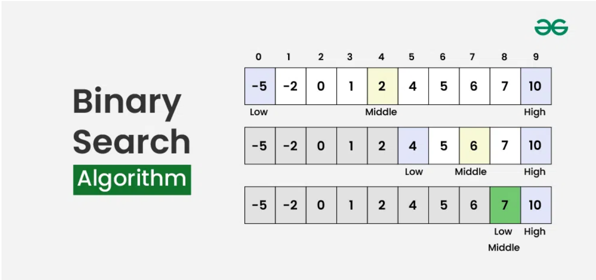
Binary search Algorithm

→ Binary search is an efficient algorithm used to find the position of a target element in a sorted list or array by repeatedly dividing the search range in half.

→ time complexity: O(log n)



To apply Binary Search algorithm:

The data structure must be sorted.
Access to any element of the data structure should take constant time.

Two pointers:

low (start of the list)

high (end of the list)

You repeatedly:

Find the middle element.

Compare it with your target.

If it matches → you're done.

If the target is smaller → search the left half.

If the target is larger → search the right half.

Keep repeating until the range collapses (low > high).

example:
Consider an array arr[] = {2, 5, 8, 12, 16, 23, 38, 56, 72, 91}, and the target = 23.